Enhancing IoT Security and Information Systems Resilience

An Extreme Value Machine Approach for Open-Set Recognition

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The Growing IoT Security Challenge

27B
IoT Devices by 2025
(Industry source: 21.1B by end 2025, 39B by 2030)

Zero-Day
Attacks Cause Most Damage
No one has seen before

Heavy-Tail
Loss Distribution
Cyber security losses follow a heavy-tail distribution. Rare events cause very large damages.

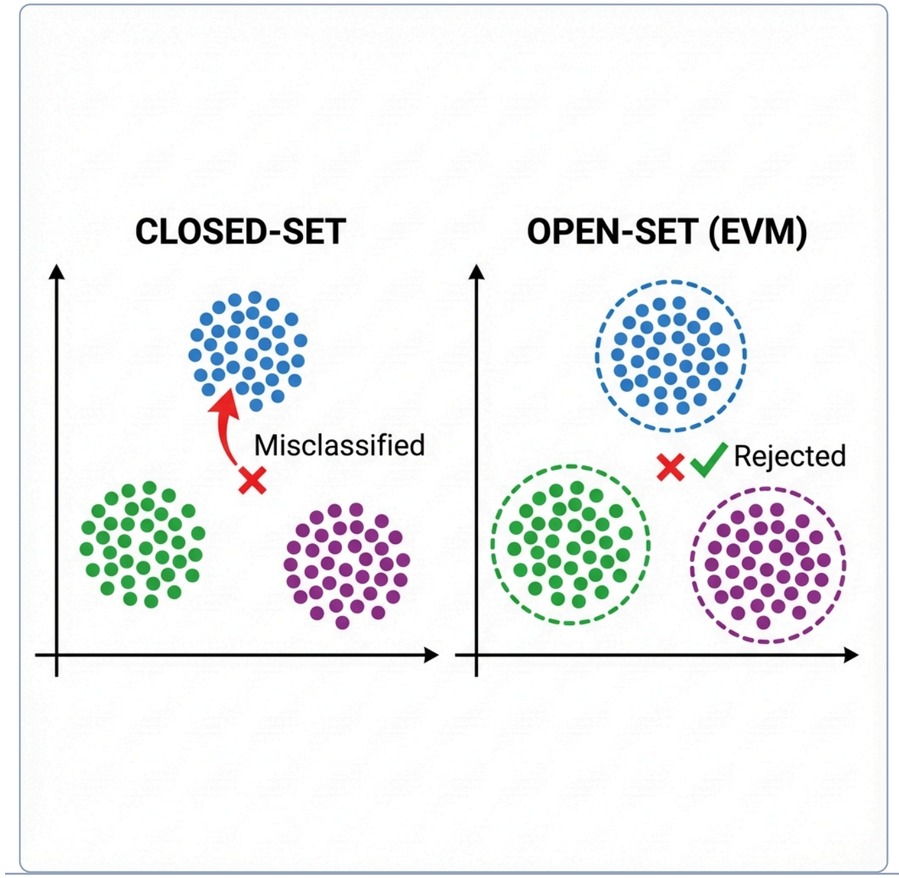
The Core Problem: Traditional intrusion detection systems work under a **closed-set assumption**. They can only detect attacks they have seen before.

Key Question: How do we protect billions of IoT devices from attacks we have never seen before?

When a new attack appears, these systems fail. This creates serious risk for organizations depending on IoT networks.

27B: Choudhary (2024). Internet of Things: Overview. Discover IoT [Academic] | 21.1B (2025), 39B (2030): IoT Analytics (2025) [Industry]

Closed-Set vs Open-Set Recognition



Closed-Set (Traditional)

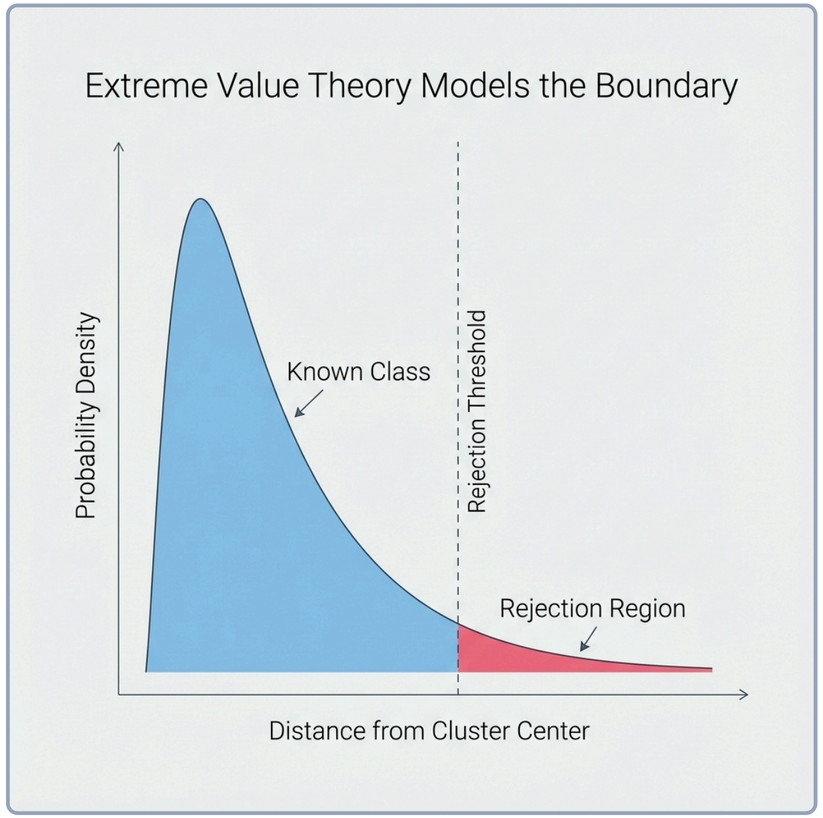
- Assumes all attack types are known
- Classifies everything into known categories
- Cannot say "I don't know this"
- Fails silently on new threats

Open-Set (Our Approach)

- Accepts that new attacks will appear
- Models boundary of known classes
- Can reject unknown samples
- Alerts on novel threats

Scheirer et al. (2013). Toward Open Set Recognition. IEEE TPAMI.

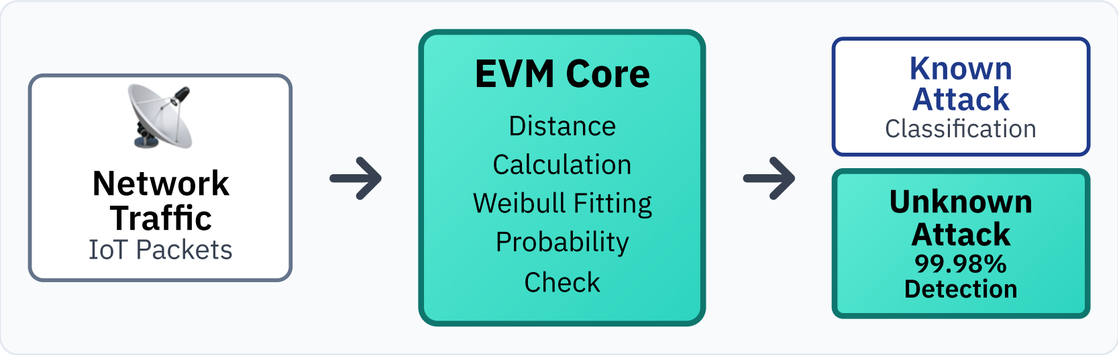
How EVM Works: Extreme Value Theory



Weibull distribution models the tail of distances.
Points beyond the threshold are rejected as unknown.

The **EVM** algorithm:

1. Calculate **distances** from new sample to known class samples
2. Fit **Weibull distribution** to extreme distances
3. Compute **inclusion probability** for each class
4. **Reject as unknown** if probability too low for ALL classes



Rudd et al. (2018). The Extreme Value Machine. IEEE TPAMI.

Research Methodology

Primary Dataset: Kitsune

- 100,000 samples (10k per class)
- 9 attack types + benign traffic
- Video Injection as "unknown"
- 70/30 train-test split

Validation Dataset: IoT-23

- 323,517 network flows
- 20 malware scenarios
- Leave-one-scenario-out protocol

Preprocessing

- Variance threshold filtering
- Standard scaling
- Feature selection (top 50)
- Unknown class held out completely

Baselines

- Isolation Forest
- One-Class SVM

Mirsky et al. (2018). Kitsune Dataset. | García et al. (2020). IoT-23 Dataset.

Results: Kitsune Dataset

99.98%

Unknown Attack Recall

Metric	EVM	Isolation Forest	One-Class SVM
Unknown Recall	**0.9998**	0.0318	0.0219
Unknown Precision	0.3227	0.1172	0.0854
Training Time	~2.1s	~0.5s	~21.7s

EVM detected 9,998 out of 10,000 unknown attack samples. Baselines detected only 2-3%.

Video Injection attack held out as unknown class.

Results: IoT-23 Dataset (Generalizability)

89.1%
Mean Unknown Recall

0.83
Mean AUROC

58.2%
Recall @ 5% FPR

Metric	Minimum	Maximum	Mean
Unknown Recall	0.564	1.000	**0.891**
Unknown Precision	0.002	0.164	0.104
Known Accuracy	0.766	0.993	0.885

20 malware scenarios tested with leave-one-out protocol.

Statistical Validation: DiD Analysis

We used **Difference-in-Differences** analysis to compare EVM against baselines across all 20 IoT-23 scenarios.

Metric	vs Isolation Forest	p-value	vs One-Class SVM	p-value
Unknown Recall	**+0.436**	$p < 0.001$	**+0.352**	$p < 0.01$
Unknown Precision	+0.262	$p < 0.001$	+0.212	$p < 0.001$
Overall Accuracy	+0.058	$p < 0.001$	+0.043	$p < 0.001$

Statistically Significant

EVM improvements are consistent across diverse malware families, not just in aggregate.

Paired t-test results with n=20 scenarios.

Ablation Studies: What Matters?

Feature Selection

Features	Recall	Accuracy
Ports only (2)	0.620	0.446
Ports + Protocol (3)	**0.963**	**0.639**
+ Duration (4)	0.963	0.600
All features (10)	0.963	0.558

Adding protocol improves recall from 62% to 96%. More features add noise.

Tail Size Parameter

Tail Size	Recall	FPR
5	0.824	0.167
10-20	**0.964**	0.510-0.555
50	0.824	0.168

Tail size controls sensitivity. Organizations can tune based on risk tolerance.

Ablation on IoT-23 dataset across 20 scenarios.

Theoretical Foundation: D&M IS Success Model

We interpret our findings through the **DeLone & McLean IS Success Model**.

System Quality

89-99%

Unknown Recall

vs 2-3% for baselines

Information Quality

32%

Precision

vs 8-12% for baselines

Net Benefits

Resilience

Prevent severe losses

Heavy-tail risk mitigation

The Trade-off: EVM's high recall means more alerts, but in IoT security, missing an attack is more costly than investigating false positives. False positives can be filtered; missed attacks cause severe damage.

DeLone & McLean (2003). D&M Model of IS Success: A Ten-Year Update. JMIS.

Practical Implications for Organizations

High-Security Environments

Use tail_size = 20 for maximum detection. Accept higher alert volume. Secondary verification for flagged traffic.

Resource-Constrained Settings

Use tail_size = 5 for lower FPR. Still achieves 82% unknown recall. Balance detection vs operational load.

Operating Point Guidance: At 5% false positive budget, EVM still recovers 58% of unknown threats. This provides concrete deployment configuration for practitioners.

Controllable trade-off based on organizational risk tolerance.

Limitations and Future Work

Current Limitations

- Computational constraints required data sampling
- Focus on offline processing
- Comparison limited to classical baselines
- Single train/test split (reproducible)

Future Directions

- Streaming EVM for real-time detection
- Benchmark against deep learning OSR methods
- Cross-validation for stability confirmation
- Integration with existing IDS infrastructure

Planned extensions to address current constraints.

Conclusion

Key Contribution

We demonstrate how **Extreme Value Theory** provides a theoretically grounded solution to open-set recognition in IoT security, achieving 89-99% unknown attack recall with statistically significant improvements over baselines.

99.98%	89.1%	p<0.001	Tunable
Kitsune Recall	IoT-23 Mean Recall	Statistical Significance	Risk Trade-off

EVM enables organizations to detect unforeseen threats and build resilient IoT security systems.

Safari & Kim (2025). Enhancing IoT Security and IS Resilience. ICIS 2025.

Thank You

Questions?

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